13.8 Bet between Stephen Hawking, John Preskill, and me on the correctness of Penrose's cosmic censorship conjecture.

Just four months after agreeing to the bet, Hawking himself discovered mathematical evidence (but *not a firm proof*) that, when a black hole completes its evaporation (Chapter 12), it might not disappear entirely as he had previously expected, but instead it might leave behind a tiny naked singularity. Hawking announced this result to Preskill and me privately, a few days after he discovered it, at a dinner party at Preskill's home. However, when Preskill and I then pressed him to concede our bet, he refused on grounds of a technicality. The wording of our bet was very clear, he insisted: The bet was restricted to naked singularities whose formation is governed by the laws of classical (that is, not quantum) physics, including the laws of general relativity. However, the evaporation of black holes is a quantum mechanical phenomenon and is governed not by the laws of classical general relativity, but rather by the laws of quantum fields in curved spacetime, so any naked singularity that might result from black-hole evaporation is outside the realm of our bet, Hawking insisted (correctly). Nevertheless, a naked singularity, however it forms, would surely be a blow to the establishment!

Though we enjoy our bets, the issues we argue are deeply serious. If naked singularities can exist, then only the ill-understood laws of quantum gravity can tell us how they behave, what they might do to spacetime in their vicinities, and whether their actions can have a large effect on the Universe in which we live, or only a small one. Because naked singularities, if they can exist, might strongly influence our Universe, we want very much to understand whether cosmic censorship is correct, and what the laws of quantum gravity predict for the behaviors of singularities. The struggle to find out will not be quick or easy.

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1. In technical jargon, we say that the singularity is "spacelike."

2. 10^{-43} second is the *Planck–Wheeler time*. It is given (approximately) by the formula $\sqrt{G\hbar/c^5}$, where G =6.670 × 10^{-8} dyne-centimeter^2/gram^2 is Newton's gravitation constant, $\hbar$= 1.055 × 10^{-27} erg-second is Planck's quantum mechanical constant, and c = 2.998 × 10^{10} centimeter/second is the speed of light. Note that the Planck-Wheeler time is equal to the square root of the Planck–Wheeler area (Chapter 12) divided by the speed of light.

3. The above description is based on the Wheeler–DeWitt, Hawking–Hartle approach to formulating the laws of quantum gravity. Although theirs is but one of many approaches now being pursued, it is